

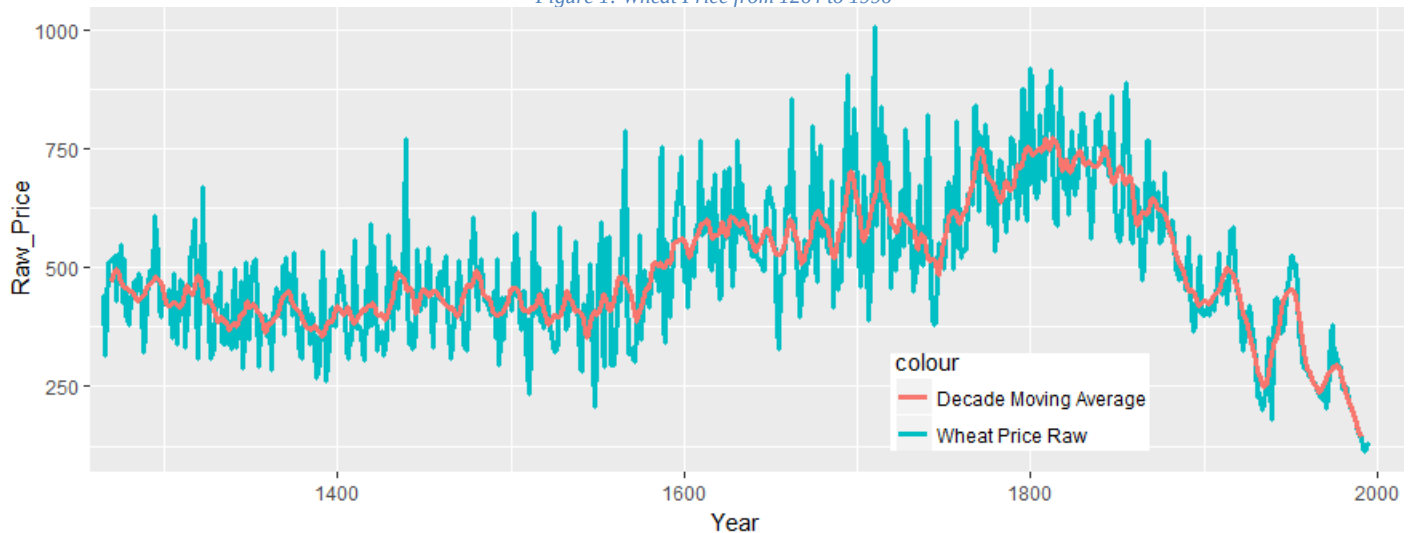
Wheat Prices through the Ages

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Introduction

This paper explores wheat prices from the year 1264 to 1996 for England in constant 1996 pounds. As with any agricultural dataset, wheat prices fluctuate seasonally. Figure 1 shows the raw prices of wheat and their 10 period moving average. Seasonality can be seen to be a lot more significant in the earlier years, when crops were usually only planted once a year. Average wheat price was about £426 till about 1590. But then from about 1590 to 1710 there is a significant upward trend in wheat price. This coincides with the general population boom, which led to increased demand for food and other commodities. This was a major driving factor behind imperial colonization. More demand in food and other commodities was a driving factor behind the industrial revolution, which happened between 1760 to about 1840. Its effects can be seen 1810 onwards, as there is a sharp decline in wheat prices. The goal of this study is to investigate the impact of technological advances on wheat, and in general on food prices. As well as to see if historical price levels are indicative of current wheat prices.

Figure 1: Wheat Price from 1264 to 1996



Modeling

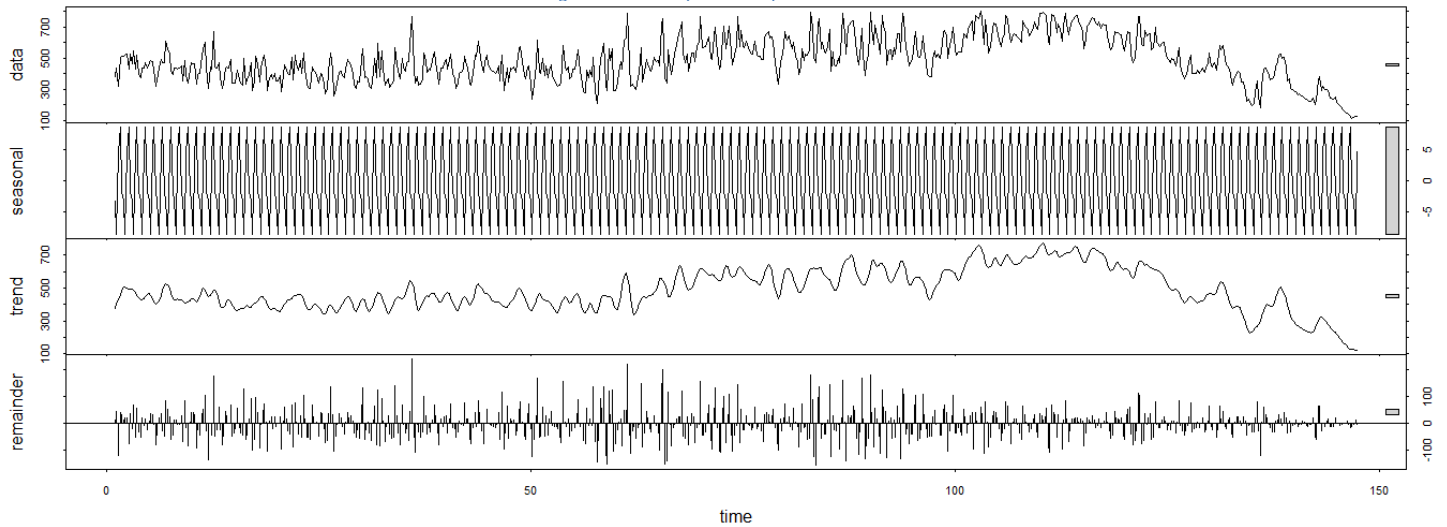
Data Cleanup

Before modeling the data was cleaned and outliers were imputed with adjacent year averages. Up until 1856 several extreme hikes can be noticed in price. We noticed 25 years ranging from 1662 to 1856 when price was increased more than 2 standard deviations. This is most likely because up until then agriculture was not heavily industrialized and food shortage was common. These outliers were toned down using price averages from adjacent years. On the other hand, the first time wheat price hit modern levels was in 1939 with £180.66. 1987 onwards wheat prices were more than 2 standard deviations below overall wheat price. But these outliers were not imputed as these indicate the modern price levels of wheat and where the direction wheat prices are headed to.

ARIMA Modeling

To fit an ARIMA model, first seasonality was removed and data was decomposed of its trend. Figure 2 shows the decomposition of the wheat data. Data has a significant seasonal trend, as well as a general trend. The remainder seems a lot more volatile during the price increasing years.

Figure 2: Decomposition of Wheat Prices

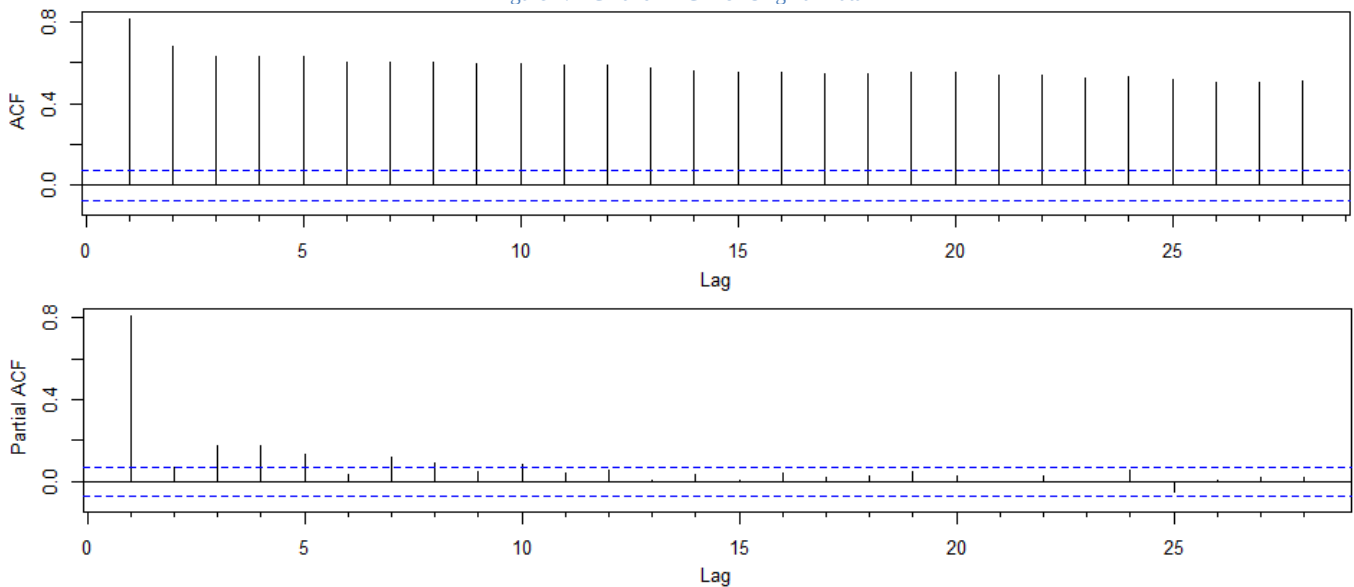


Second, we check stationarity of the data. We know there's a strong trend present, so data is not expected to be stationary. This is why an ARIMA model is decided to be used. Figure 3 shows the p-value for the Augmented Dickey-Fuller test is 0.796, so we cannot reject the null-hypothesis that the data is non-stationary. Figure 4 shows a slow drop in ACF followed by a sudden cutoff in PACF, which is indicative of an AR(1) process.

Figure 3: Original Data Stationarity Test
Augmented Dickey-Fuller Test

```
data: wheat$Price
Dickey-Fuller = -1.4857, Lag order = 9, p-value = 0.796
alternative hypothesis: stationary
```

Figure 4: ACF and PACF for Original Data



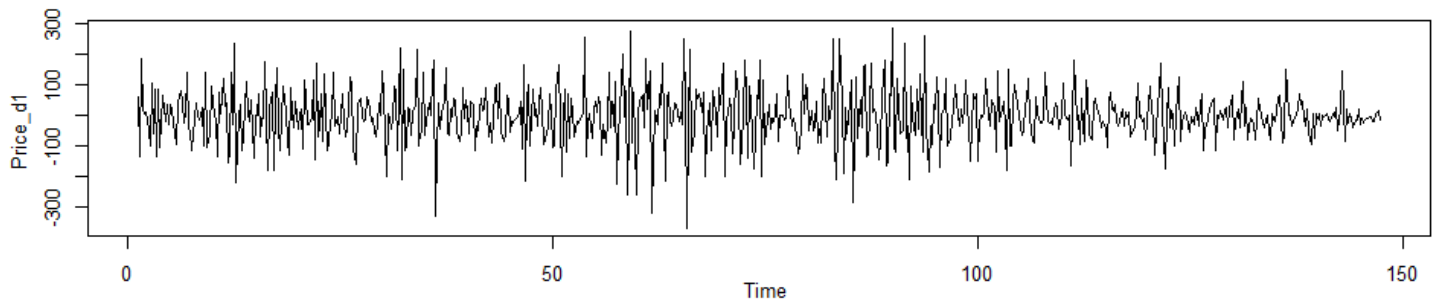
In order to fit an ARIMA model, the first step was to determine the difference which would make the data stationary. In this case, the first difference achieves stationarity with a p-value of 0.01 (Figure 5) for the Augmented Dickey-Fuller Test.

Figure 5: First Difference Stationarity Test
Augmented Dickey-Fuller Test

```
data: Price_d1
Dickey-Fuller = -13.638, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary
```

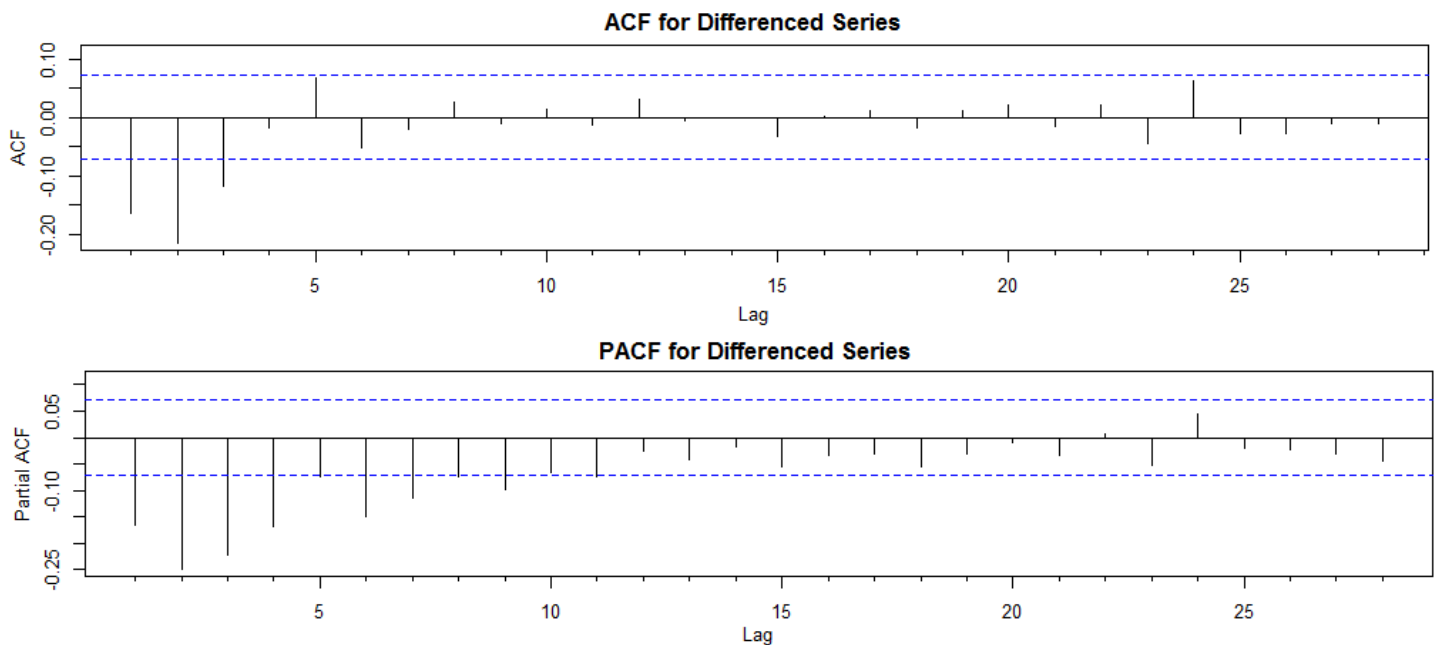
Figure 6 shows the plot for the differenced data. It appears to have balanced variance about 0, but there is some volatility in the data towards the middle.

Figure 6: First Difference Plot



Next, ACF and PACF is re-examined as shown in Figure 7. ACF shows that lags 1, 2, and 3 are significant. PACF has several significant lags up until lag 9. Several significant lags suggest $p>1$ and/or $q>1$ for our ARIMA model. All significant lags are negative, which suggests $\theta<0$ and $\phi>0$.

Figure 7: ACF and PACF for First Difference



We utilize the auto ARIMA function in R to find the best fit for our data. The results can be seen in Figure 8. Auto ARIMA suggests $p=1$, $d=1$ and $q=2$. The residuals plot shows some slightly significant lags after lag 30. This confirms that our data has an AR(1) tendency. However, the residuals in Figure 9 show some clustered significant lags between lag 30 and 40.

Figure 8: ARIMA Fit

```
Series: deseasonal_val
ARIMA(1,1,2)

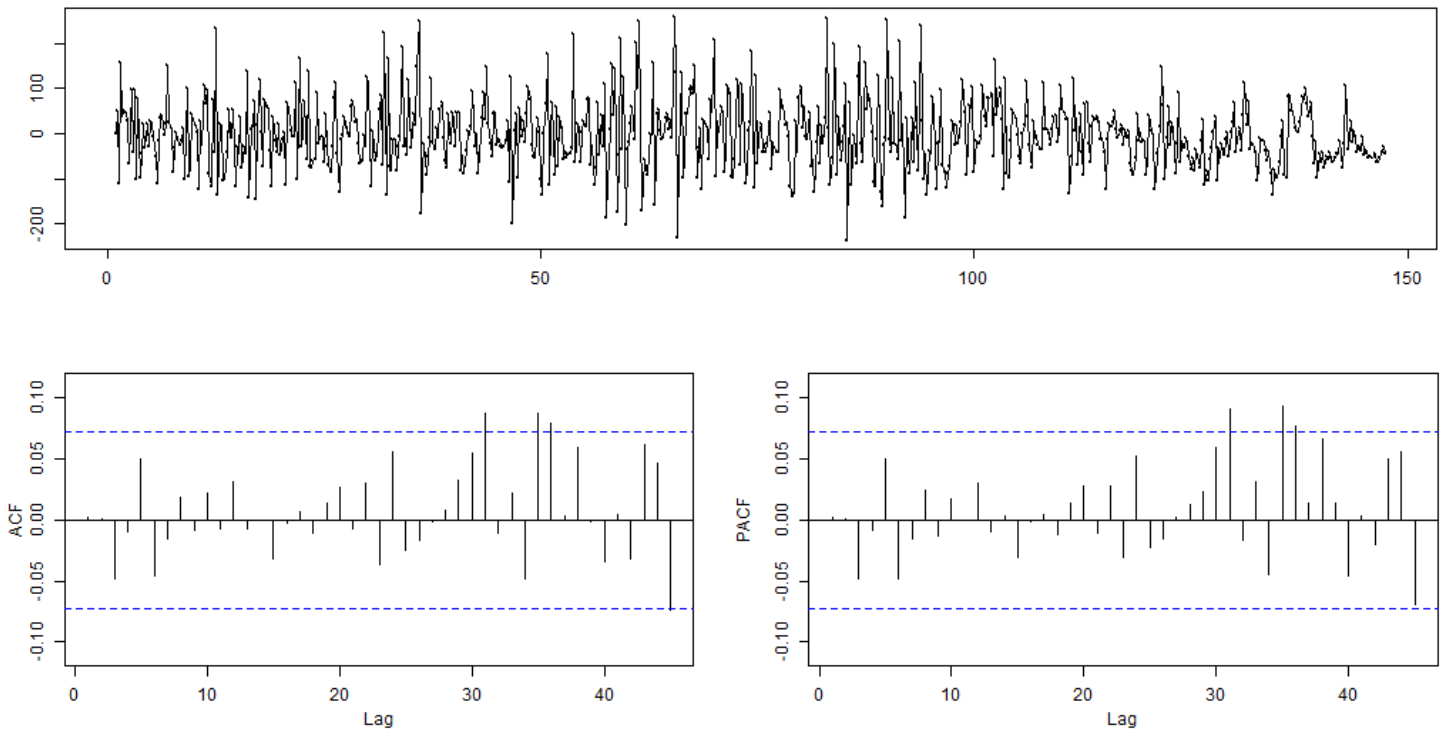
Coefficients:
      ar1      ma1      ma2
    0.2801  -0.6677  -0.2206
s.e.  0.0775   0.0766   0.0633

sigma^2 estimated as 5962:  log likelihood=-4219.37
AIC=8446.74  AICC=8446.8  BIC=8465.12
```

Fitted Model Equation for first difference:

$$Price_{d=1} = 0.2801t_{-1}\phi - 0.6677t_{-1}\theta - 0.2206t_{-2}\theta + E$$

Figure 9: ARIMA(1,1,2) Residuals
(1,1,2) Model Residuals



To further refine this model, we used the SARIMA function available in R. As shown in Figure 10, this added a constant term to our ARIMA(1,1,2) model, which slightly reduces the significant lags as shown in Figure 11.

Figure 10: SARIMA(1,1,2) Model Fit

```

Coefficients:
      ar1      ma1      ma2  constant
    0.2760 -0.6658 -0.2239  -0.3759
s.e.  0.0774  0.0762  0.0633   0.4413

sigma^2 estimated as 5972:  log likelihood = -4221.51,  aic = 8453.02

$degrees_of_freedom
[1] 729

$table
      Estimate      SE t.value p.value
ar1      0.2760 0.0774  3.5677  0.0004
ma1     -0.6658 0.0762 -8.7317  0.0000
ma2     -0.2239 0.0633 -3.5364  0.0004
constant -0.3759 0.4413 -0.8517  0.3946

$AIC      $AICC      $BIC
[1] 9.705766 [1] 9.708607 [1] 8.730852

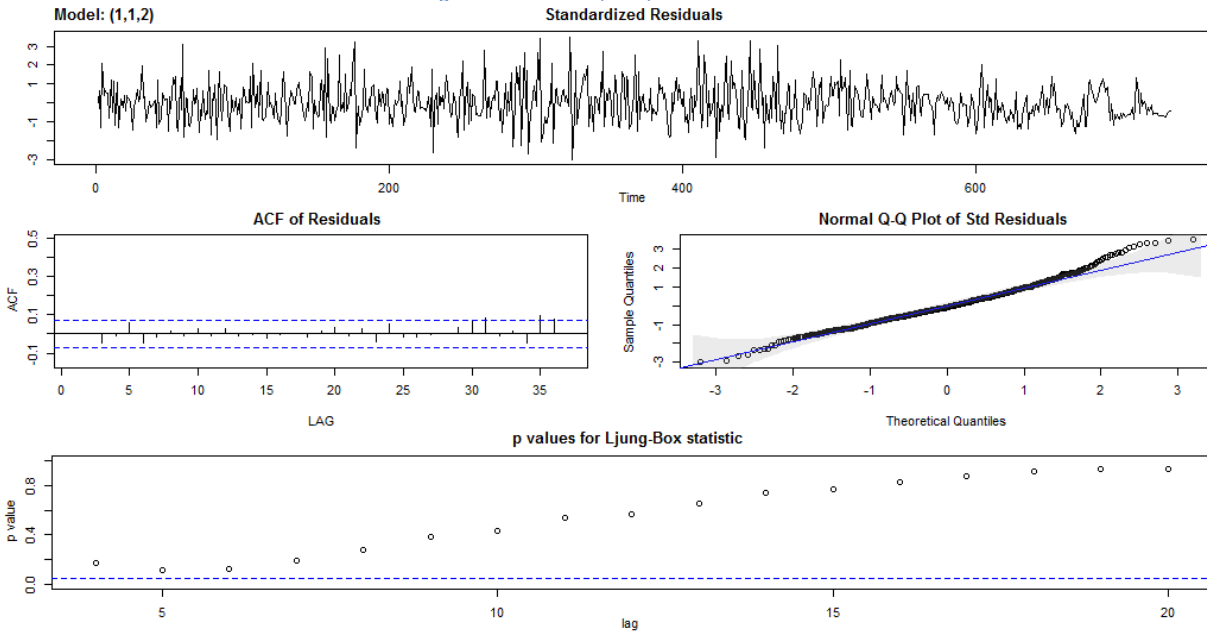
```

The revised model equation for first difference is:

$$Price_{d=1} = -0.3759 + 0.2760_{t-1}\varphi - 0.6658_{t-1}\theta - 0.2239_{t-2}\theta + E$$

Furthermore, the Q-Q plot indicates two outliers beyond ± 3 . All p-values for the Ljung-Box test are insignificant, which indicate that model residuals are independent.

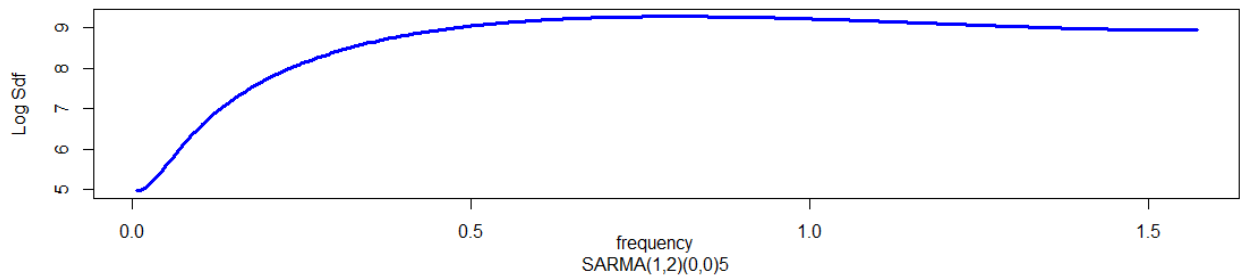
Figure 11: SARIMA(1,1,2) Model Residuals



Spectral Density of Model

Increasing spectral density till 0.5 suggests AR(1). The decreasing spectral density after 0.5 suggests $\theta < 0$, as already observed in fitted model.

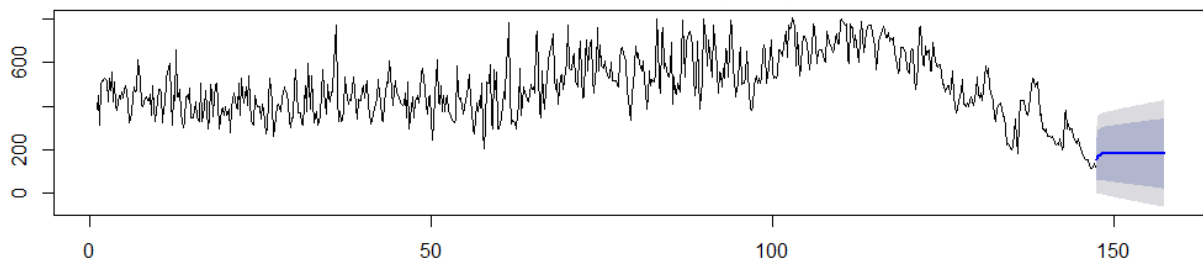
Figure 12: Spectral Density of Fitted Model



Conclusion

Finally, we look at how our model forecasts the next 50 periods. The forecast predicts a slight increase, and then flats out at right below £200. Which is about £350 today. However, the actual wheat price in the UK today is about £175. This indicates that the downward trend in wheat prices continued on much longer than our seasonal ARIMA model anticipated. This is likely due to the fact that the decline was only apparent in the last 200 years, whereas the data spans 733 years. The failure of the model to predict current wheat price correctly indicates that further advances in technology have accelerated the decline in wheat price more than it was anticipated back in the 90's.

Figure 13: 50 Period Forecast
Forecasts from ARIMA(1,1,2)



Data and R Code Used

Data was obtained from: https://datamarket.com/data/set/22ln/wheat-prices-1264-1996-in-constant-1996-pounds?_escaped_fragment_=%3D22ln%26display%3Dline#!ds=22ln&display=line

R Code

```
#Analysis
```

```
decomp = stl(Wheat$Price)
deseasonal_val <- seasadj(decomp)
plot(decomp)
```

```
#Checking stationarity
```

```
adf.test(Wheat$Price, alternative = "stationary")
win.graph(width=5,height=3,pointsize=8)
par(mfrow=c(2,1))
Acf(Wheat$Price, main="")
Pacf(Wheat$Price, main="")
```

```
#Finding correct difference
```

```
Price_d1 = diff(deseasonal_val, differences = 1)
plot(Price_d1)
adf.test(Price_d1, alternative = "stationary")
```

```
#Selecting p and q
```

```
win.graph(width=5,height=3,pointsize=8)
par(mfrow=c(2,1))
Acf(Price_d1, main='ACF for Differenced Series')
Pacf(Price_d1, main='PACF for Differenced Series')
```

```
#Residuals
```

```
fit<-auto.arima(deseasonal_val, seasonal=FALSE)
fit
tsdisplay(residuals(fit), lag.max=45, main='(1,1,2) Model Residuals')
```

```
#Forecast
```

```
fcast <- forecast(fit, h=50)
plot(fcast)
```

```
#SARIMA
```

```
fit2<-sarima(Wheat$Price, 1, 1, 2, P = 0, D = 0, Q = 0, S = -1, details = TRUE, xreg=NULL, Model=TRUE, tol = sqrt(.Machine$double.eps), no.constant = FALSE)
fit2
fcast <- forecast(fit2, h=50)
plot(fcast)
```

```
#spectral
```

```
sdfplot(fit2)
```